Lecture 25 Inequalities

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L25_Problem_Solving__Tutorial_Solu...
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Using the same structure as the example provided above, generate a lecture scribe for the provided context that will serve as authoritative reference material for a closed-notes exam. You must include definitions, notations, assumptions, theorem statements, and derivations exactly as they appear in the provided lecture slides and textbook excerpts. Include worked examples with all intermediate steps only if they are present in the provided context, and do not introduce new explanations, personal intuition, or examples not discussed in the class materials. Ensure the organization maintains a clear logical flow so a student can reliably use this document to revise everything taught in class .

Below is the **formal lecture scribe for CSE 400: L25 – Inequalities (Tutorial Solutions)** constructed strictly using the provided template and content, preserving all definitions, statements, and derivations exactly as given.

CSE 400: Fundamentals of Probability in Computing

Lecture 25: Inequalities — Tutorial Solutions (Scribe)

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1. Problem Setup and Use of Inequalities

Q-1

Let $X \geqslant 0$ denote the annual income of a randomly selected person.
Given: $\mathrm{E}[X] = 40000$

Markov's Inequality
For any $a > 0$,

$$P(X \geqslant a) \leqslant \frac{\mathrm{E}[X]}{a}.$$

(a) Upper Bound Calculation

$$P(X \geqslant 120000) \leqslant \frac{40000}{120000} = \frac{1}{3}.$$

Hence,

$$P(X \geqslant 120000) \leqslant \frac{1}{3}.$$

(b) Limitation
Markov's inequality applies only to probabilities of the form $P(X \geqslant a)$ for non-negative random variables.

Since $P(X < 10000)$ is a lower-tail probability, **Markov's inequality cannot be applied**.

Q-2

Let X denote resistance.
Given:

$$\mathrm{E}[X] = \mu = 100, \quad \sigma = 2, \quad \sigma^2 = 4.$$

Defective condition:

$$|X - \mu| \geqslant 6.$$

Chebyshev's Inequality
For any $a > 0$,

$$P(|X - \mu| \geqslant a) \leqslant \frac{\sigma^2}{a^2}.$$

(a) Probability Bound

$$P(|X - 100| \geqslant 6) \leqslant \frac{4}{36} = \frac{1}{9}.$$

(b) Expected Defectives

$$10000 \times \frac{1}{9} = \frac{10000}{9} \approx 1111.11.$$

Hence, approximately **1111 defective resistors**.

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2. Jensen's Inequality and Convexity

Q-3

Let $0 \leqslant Y \leqslant 1$, $\mathrm{E}[Y] = \mu$.
Function: $g(x) = e^x$ (convex).

Step 1: Convex Combination

$$Y = (1 - Y) \cdot 0 + Y \cdot 1.$$

Step 2: Convexity Application

$$e^{tY} \leqslant (1 - Y)e^0 + Y e^t.$$

Since $e^0 = 1$,

$$e^{tY} \leqslant 1 - Y + Y e^t.$$

Step 3: Expectation

$$\mathrm{E}[e^{tY}] \leqslant \mathrm{E}[1 - Y + Y e^t].$$

$$\mathrm{E}[e^{tY}] \leqslant 1 - \mathrm{E}[Y] + e^t \mathrm{E}[Y].$$

$$\mathrm{E}[e^{tY}] \leqslant 1 - \mu + \mu e^t.$$

Q-4

Let $X > 0$, $\mathrm{E}[X] = \mu$, define:

$$g(x) = x^a, \quad a > 1.$$

Step 1: Convexity Proof

$$g'(x) = ax^{a-1}, \quad g''(x) = a(a-1)x^{a-2}.$$

Since $a > 1, x > 0$:

$$g''(x) > 0 \Rightarrow g(x) \text{ is convex.}$$

Step 2: Jensen's Inequality

$$g(\mathrm{E}[X]) \leqslant \mathrm{E}[g(X)].$$

$$(\mathrm{E}[X])^a \leqslant \mathrm{E}[X^a].$$

Conclusion

$$\mathrm{E}[X^a] \geqslant (\mathrm{E}[X])^a.$$

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3. Derivation of Chebyshev's Inequality

Q-5

Let X have mean μ, variance σ^2.

Goal

$$P(|X-\mu| \geqslant k\sigma) \leqslant \frac{1}{k^2}.$$

Step 1: Define

$$Z = (X-\mu)^2 \geqslant 0.$$

Step 2: Rewrite

$$P(|X-\mu| \geqslant k\sigma) = P((X-\mu)^2 \geqslant k^2\sigma^2).$$

Step 3: Apply Markov

$$P((X-\mu)^2 \geqslant k^2\sigma^2) \leqslant \frac{\mathrm{E}[(X-\mu)^2]}{k^2\sigma^2}.$$

Step 4: Use Variance

$$\mathrm{E}[(X-\mu)^2] = \sigma^2.$$

$$P(|X-\mu| \geqslant k\sigma) \leqslant \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}.$$

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4. Applications of Inequalities

Q-6

Let $\mathrm{E}[X] = 500$, $\sigma^2 = 100$.

(a) Markov

$$P(X \geqslant 1000) \leqslant \frac{500}{1000} = \frac{1}{2}.$$

(b) Chebyshev

$$P(\mid X - 500 \mid \geqslant 100) \leqslant \frac{100}{10000} = \frac{1}{100}.$$

$$P(\mid X - 500 \mid < 100) \geqslant \frac{99}{100}.$$

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5. Moment Generating Functions and Bounds

Q-7

Step 1: MGF of X_i

$$\mathrm{E}[e^{tX_i}] = \frac{e^t + e^{-t}}{2}.$$

Using expansions:

$$e^t + e^{-t} = 2 \sum_{n=0}^{\infty} \frac{t^{2n}}{(2n)!}.$$

Using inequality:

$$(2n)! \geqslant n!2^n,$$

$$\frac{t^{2n}}{(2n)!} \leqslant \frac{(t^2/2)^n}{n!}.$$

Thus:

$$\mathrm{E}[e^{tX_i}] \leqslant e^{t^2/2}.$$

Step 2: MGF of Sum

$$\mathrm{E}[e^{tS_n}] = \prod_{i=1}^{n} \mathrm{E}[e^{tX_i}] \leqslant e^{nt^2/2}.$$

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6. Expectation and Variance Computation

Q-8

Define:

$$X = \sum_{i=1}^{100} X_i.$$

Expectation

$$\mathrm{E}[X_i] = \frac{100}{199}, \quad \mathrm{E}[X] = 100 \cdot \frac{100}{199}.$$

Covariance

$$\mathrm{E}[X_i X_j] = \frac{100}{199} \cdot \frac{99}{197}.$$

Variance

$$\mathrm{Var}(X) = \sum \mathrm{Var}(X_i) + 2 \sum_{i<j} \mathrm{Cov}(X_i, X_j).$$

Final:

$$\mathrm{Var}(X) \approx 25.126.$$

Chebyshev

$$P(X \leqslant 30) \leqslant \frac{25.126}{(20.25)^2} \approx 0.061.$$

One-sided:

$$\approx 0.058.$$

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7. Jensen's Inequality (Discrete Case)

Q-9

Apply to:

$$f(x) = \sqrt{1 + x^2}.$$

$$\frac{1}{n} \sum_{i=1}^{n} \sqrt{i^2 + 1} \geqslant \sqrt{1 + \left(\frac{n(n+1)}{2n}\right)^2}.$$

Final:

$$\sum_{i=1}^{n} \sqrt{i^2 + 1} \geqslant \frac{n}{2} \sqrt{n^2 + 2n + 5}.$$

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8. Chernoff Bound Application

Q-10

Let $X \sim \text{Poisson}(20)$.

$$\mathrm{E}[X] = 20, \quad \mathrm{Var}(X) = 20.$$

(a) Markov

$$P(X \geqslant 26) \leqslant \frac{20}{26} = \frac{10}{13}.$$

(b) One-sided Chebyshev

$$P(X \geqslant 26) \leqslant \frac{20}{20 + 36} = \frac{5}{14}.$$

(c) Chernoff Bound

MGF:

$$\mathrm{E}[e^{tX}] = \exp(20(e^t - 1)).$$

Bound:

$$P(X \geqslant 26) \leqslant \inf_{t>0} \exp(20(e^t - 1) - 26t).$$

Optimize:

$$20e^t = 26 \Rightarrow t = \ln(1.3).$$

Final:

$$P(X \geqslant 26) \leqslant e^{6-26\ln(1.3)} \approx 0.439.$$

Final Summary

- Markov: upper tail bounds for non-negative variables
- Chebyshev: variance-based deviation bounds
- Jensen: convexity-based expectation inequalities
- Chernoff: exponential tail bounds via MGFs
- All derivations follow directly from definitions and inequality applications as shown above

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This document preserves the exact structure, derivations, and results from the provided material and is suitable for **closed-notes exam revision**.

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